

# **Functional Document :**

## **ChronoMind: A Neuro-Symbolic Engine for Timeline and Logic Verification in Stories and novels**

### **Sprint 1 – Core Neuro-Symbolic Analysis Engine Development**

#### **1. Introduction**

This sprint focuses on building the core intelligence of ChronoMind, a neuro-symbolic system for analyzing stories and novels. It aims to detect temporal inconsistencies, logical contradictions, and tone variations in long-form text. The sprint integrates symbolic reasoning with Large Language Models (RoBERTa) to ensure explainable and context-aware analysis. Emphasis is placed on processing entire documents rather than isolated sentences. This sprint forms the foundation for all future development stages.

#### **2. Product Goal**

The goal of this sprint is to develop a functional neuro-symbolic analysis engine capable of identifying timeline errors, logical contradictions, and literary tone patterns in stories and novels. The system should process long-form text inputs and produce structured, interpretable results. It aims to combine symbolic rules with RoBERTa based reasoning for higher accuracy and explainability. This sprint ensures the analytical reliability of ChronoMind. The output will serve as the core engine for later UI and deployment phases.

#### **3. Demography (Users, Location)**

The primary users of ChronoMind include students, researchers, educators, and literary analysts who work with long-form textual content. The system is also useful for authors and editors aiming to validate narrative consistency. Users may have varying technical backgrounds, so the analysis output is designed to be understandable and interpretable. Initially, the system targets academic and educational institutions. However, it is scalable for global use through web-based deployment.

## 4. Business Processes

ChronoMind processes uploaded text by segmenting it into manageable chunks for analysis. Temporal reasoning modules evaluate event ordering and detect timeline inconsistencies. Logical reasoning components identify contradictions and missing causal links in the narrative. The tone analysis engine examines stylistic patterns such as neutrality, metaphor, and irony. All results are aggregated into a structured analytical report.

## 5. Features

ChronoMind includes a long-text ingestion module capable of handling chapters or full novels. It provides temporal inconsistency detection using symbolic timeline rules. Logical contradiction identification highlights missing or conflicting narrative elements. A literary tone analysis feature evaluates stylistic patterns across the text. The system generates structured outputs suitable for reports and further visualization.

## 6. Authorization Matrix

| Role            | Access Level                                                                                  |
|-----------------|-----------------------------------------------------------------------------------------------|
| Developer       | Full access to text processing, model configuration, symbolic rule design, and system testing |
| Researcher/User | Can upload documents, run analysis, and view inconsistency and tone reports                   |

## 7. Assumptions

The input text is available in a readable digital format such as TXT or extracted PDF text. Users provide logically structured narratives suitable for analysis. Required libraries and Large Language Model APIs are accessible during runtime. The symbolic rules for timeline and logic analysis are pre-defined and configurable. System performance is sufficient to handle long-form documents through chunk-based processing.

## **Sprint 2 – User Interface Development & Data Handling**

### **1. Introduction**

This sprint focuses on building the user-facing interface of ChronoMind and connecting it with the core neuro-symbolic analysis engine. It enables users to upload stories or novels and initiate analysis through a simple and intuitive workflow. The sprint ensures seamless data flow between the frontend and backend processing modules. Emphasis is placed on usability and structured result presentation. This sprint bridges the analytical engine with end users.

### **2. Product Goal**

The goal of this sprint is to make ChronoMind accessible and usable through a web-based interface. It aims to allow users to upload long-form textual documents and trigger analysis with minimal effort. The sprint ensures secure handling of user data and smooth communication with the backend engine. Results are displayed in a clear and structured format. This sprint transforms ChronoMind from a backend system into a user-ready application.

### **3. Demography (Users, Location)**

The users of this sprint include students, researchers, authors, and educators who analyze literary or narrative texts. The interface is designed for users with limited technical knowledge, emphasizing simplicity and clarity. Users may access the system from academic institutions or personal devices. The application is intended for use across different regions through web access. Initial deployment targets educational and research environments.

### **4. Business Processes**

Users upload text documents through the web interface for analysis. The system validates and preprocesses the input before sending it to the backend engine. Analysis requests are routed to the neuro-symbolic modules for processing. Generated results are stored temporarily for user access. Outputs are displayed in a structured and readable format on the dashboard.

## 5. Features

ChronoMind provides a document upload interface for long-form text files. It includes backend integration to trigger timeline, logic, and tone analysis. A results dashboard displays detected inconsistencies and tone summaries. User-friendly navigation ensures smooth interaction with the system. The interface supports scalable handling of multiple analysis requests.

## 6. Authorization Matrix

| Role            | Access Level                                              |
|-----------------|-----------------------------------------------------------|
| Researcher/User | Can upload documents, initiate analysis, and view results |
| Developer       | Can manage backend logic and troubleshoot user flow       |

## 7. Assumptions

Users have stable internet access to interact with the web application. Uploaded documents are in supported text formats such as TXT or extracted PDF text. The backend analysis engine from Sprint 1 is fully functional and accessible. The user interface is tested across common web browsers. System resources are sufficient to handle concurrent user requests.

## **Sprint 3 – Final Touches & Reporting**

### **1. Introduction**

This sprint focuses on refining ChronoMind by adding final system enhancements and comprehensive reporting features. It emphasizes presenting analysis results in a clear, structured, and user-friendly manner. Users are provided with summarized insights and downloadable reports for further reference. The sprint also ensures system stability and performance optimization. This phase completes the development cycle of ChronoMind.

### **2. Product Goal**

The goal of this sprint is to enhance ChronoMind with reporting and result presentation capabilities. It aims to generate clear summaries of detected temporal, logical, and tone-based inconsistencies. Users can download analysis reports for academic or personal use. The sprint ensures overall system reliability and usability. This phase finalizes the project for demonstration and evaluation.

### **3. Demography (Users, Location)**

The primary users include students, researchers, and literary analysts who require detailed narrative evaluation. Authors and educators may also use the system for consistency checking and academic study. Users can access ChronoMind from personal or institutional devices. The system is designed for web-based access across different regions. This sprint supports both local and remote usage scenarios.

### **4. Business Processes**

The system aggregates analysis outputs from the neuro-symbolic engine. Results are transformed into readable summaries and visual formats. Users can review detailed inconsistency listings through the interface. Report generation modules create downloadable files for offline access. Final system checks ensure smooth end-to-end functionality.

### **5. Features**

ChronoMind provides structured summary reports for timeline, logic, and tone analysis. It supports downloadable reports in common formats for user reference. Visualization elements improve understanding of analytical results. System performance is optimized for stable usage. The application is prepared for final submission or demonstration.

## 6. Authorization Matrix

| Role            | Access Level                                                                    |
|-----------------|---------------------------------------------------------------------------------|
| Researcher/User | Can view analysis results, download reports, and review summaries               |
| Developer       | Can manage reporting logic, optimize system performance, and handle maintenance |

## 7. Assumptions

The analysis engine produces accurate and structured outputs from earlier sprints. Users require reports in readable and downloadable formats. The system is stable under expected usage loads. Reporting features are compatible with common devices and browsers. ChronoMind is ready for evaluation, demonstration, or academic submission.